



Food shortage affects reproduction of Feral Pigeons *Columba livia* at rearing of nestlings

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Feral Pigeons *Columba livia* are highly adapted to urban environments and are thus often abundant in cities. This can lead to various problems, including fouling of building facades and pavements, transmission of allergens and pathogenic micro-organisms, and infestations of ectoparasites derived from breeding sites. To develop effective, long-lasting and humane control strategies, it is necessary to understand the demography of Feral Pigeons. Although food shortage is a major source of reproductive failure in Feral Pigeons, it is still unclear at which phase of the reproductive cycle this reduces overall reproductive success. Here, we assess the effect of a sudden reduction in the food base on the reproduction of a well-studied Feral Pigeon breeding colony. The findings of this study suggest that the number of broods per pair decreases significantly during food scarcity, and that although hatching success remains constant, a significantly greater number of nestlings die during the rearing phase. This suggests that the high energy demand of Feral Pigeon nestlings could not be met under conditions of food scarcity, which reduced the total number of fledged young by more than half and led to a reduction in the colony size. These results have important implications for selecting suitable, durable and humane control strategies for the management of large Feral Pigeon populations in urban environments.



Keywords: breeding biology, food loss, food reduction, nestling mortality, nutrition, population regulation.



INTRODUCTION

Food supply limits reproductive success and survival of both the young and the parents in many bird species (Martin 1987). For example, artificially reduced brood sizes or increased food supply result in a consistent increase in size and survival prospects of the young (Martin 1987).

Feral Pigeons are the free-living descendants of domesticated forms of the wild Rock Dove *Columba livia*. They populate most cities in temperate regions (Baptista *et al.* 1997). The size of Feral Pigeon populations is mainly determined by their food supply. The available food base is intentionally supplied by humans or available as human food waste. The partial absence of effective

regulation by predators and the rich food base allow the development and maintenance of large populations that can cause human health problems, including fouling with faeces and transmission of pathogenic micro-organisms and parasites to humans (Haag 1991). Each pigeon produces 4–11 kg of faeces annually (Vogel 1997), soiling breeding areas, house facades, monuments, streets and other city sites. Zucconi *et al.* (2003) estimated the overall damage to be €23.7–33.5 (US\$ 27–38) per Feral Pigeon per year. Additionally, 111 pathogenic agents and 20 harmful arthropod species that can infest humans have so far been found in Feral Pigeon populations (Haag-Wackernagel & Moch 2004, Haag-Wackernagel & Bircher 2010). While many wild-living species have a parasitic fauna comparable to that of Feral Pigeons, no other species lives as close to humans and offers so many possibilities of transmission. For all these

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reasons, the Feral Pigeon represents an important study species for which to consider the impact and mechanisms of changing food supply on reproduction.

Estimates of pigeon demographic parameters and their variability are crucial when selecting a suitable control strategy because they provide a working basis for determining the feasibility of deterrence or population control (Giunchi *et al.* 2012). Principal options include the protection of specific buildings or city-wide solutions aimed at the reduction of pigeon populations. However, most interventions either lack scientific proof of efficacy or have failed to prove effective when tested (Stock & Haag-Wackernagel 2014). Furthermore, several studies have demonstrated that culling has no long-lasting effect on the population size, because Feral Pigeons tend to fill the losses quickly through compensatory natality and immigration (Murton *et al.* 1972, Haag 1984, Kautz & Malecki 1990, Sol & Senar 1995, Schnitzler 1999).

Many birds adjust their clutch size to environmental conditions, having smaller clutches when food is scarce and larger clutches when it is abundant. This ability to vary clutch size is one way of allocating reproductive energy (Burley 1980). Another option is to modify the interval between successive clutches, thus creating the possibility in good food conditions of clutch overlap where two successive clutches are being cared for simultaneously (Burley 1980). Feral Pigeons pursue a breeding strategy that aims to achieve the greatest number of broods to produce as many fledglings as possible (Hetmański & Barkowska 2008). Relevant evolutionary strategies include two-egg clutches, small eggs (Robertson 1988), quick replacement of lost clutches (Johnston & Janiga 1995), feeding the nestlings with highly nutritious crop milk (Gillespie *et al.* 2012), overlapping clutches (Burley 1980, Hetmański & Wołk 2005), biparental care of the brood and all-season breeding under optimal feeding conditions (Häkkinen *et al.* 1973, Johnston & Janiga 1995).

The raising of young is time- and energy-consuming for both parents. During the first few days, nestlings are fed exclusively with crop milk. When the squabs grow older, the milk is mixed with soaked grains and is gradually replaced by grains only (Vandeputte-Poma 1980). As a result of being fed highly nutritious crop milk, pigeon squabs have a uniquely high growth rate. There

is no parallel among the young of animal species studied so far to the 22-fold increase in body weight in the first three postnatal weeks seen in pigeon squabs (Shetty *et al.* 1992). Biparental feeding of crop milk allows Pigeons to raise their young independently of specific nestling food. The feeding of the quickly developing young, however, comes with an increased energy input and enormous resource requirements for both parents. An adult rearing a 10-day-old nestling consumes 243% of the average daily food intake of a non-incubating single adult (Riddle & Braucher 1934).

Feral Pigeon populations depend heavily on abundant food to pursue these reproductive strategies (Haag 1984). A reduction in food supply leads to increased investment in foraging, which in turn reduces reproductive output, and could reduce population size if neither immigration nor survival rates are able to compensate. While the impact of food limitation on Feral Pigeon populations has already been described (Johnston & Janiga 1995), it is unknown at which phase of the reproductive cycle a food shortage reduces overall reproductive success. In this study we investigated the effect of a reduction in the food base on the reproductive cycle of a well-studied Feral Pigeon breeding flock under natural living conditions over a period of 8 years. We predicted that the number of Feral Pigeon breeding pairs and broods per pair would be limited under food shortage. We also expected the interval between clutches to increase. Additionally, as some species limit their clutch size under less favourable environmental conditions, the number of eggs per brood provides insights into alternative reproductive strategies in adapting to a limited food base. We thus tested whether the number of eggs per clutch and per year was reduced after food limitation took place. Because the Pigeons had to spend more time foraging, we also expected reduced hatching and fledging success. Lastly, we examined whether any change in reproductive output led to an overall change in population size after food limitation. Our overall aim was to define the extent and main drivers of any change in reproduction under conditions of food scarcity. Specialists dealing with pest species depend on such results when choosing the most promising and durable control strategy against an abundant species, such as the Feral Pigeon, and to implement such strategies to achieve an

appropriate and humane existence of Feral Pigeon populations in urban environments.

METHODS

Study site

The Feral Pigeon population used for this study occupies a pigeon loft managed by the authors within the framework of the 'Pigeon Action of Basel' (Haag-Wackernagel 1993). The loft is situated in the St Matthew Church in a residential quarter on the outskirts of Basel, Switzerland (47°34'N, 7°35'E). It has a floor space of 28 m² and is equipped with 39 breeding boxes, although a few pairs also nest on the floor or on ledges formed by beams and boards. The loft is cleaned of droppings, nesting material and carcasses every alternate week, and breeding data as well as general observations are recorded. The Pigeons use the loft as a roosting and breeding site only. They are free to enter or leave at any time with the exception of nocturnal censuses, during which all Pigeons present are captured and data, as specified below, are recorded. Pigeons hatched in the loft are individually marked with an aluminium leg ring at fledging age, and immigrant birds are ringed during the nocturnal censuses. Every ringed Pigeon has been registered in a database since the opening of the loft. The colony varies seasonally between 75 and 100 individuals. The birds are not fed by the researchers and must search for food and water themselves. Thus, they are exposed to the prevailing urban conditions. In a previous study (Rose *et al.* 2006) we found that the birds foraged mainly within 300–2000 m of the loft, and especially at the Rhine Port St Johann, 1 km away, where large amounts of wasted grain were consistently available. The port management have stated that approximately 60 tons of grain were spilled yearly, most of which was consumed by Feral Pigeons. Assuming a daily intake of 29 g per individual (Haag 1984), this food source alone could feed around 5700 Feral Pigeons throughout the year. Additionally, the birds from the study loft were recorded foraging in local streets and other areas around the loft. At the end of December 2009, the Rhine Port St Johann was closed and demolished. The population thus lost its main food source within a few days. This allowed us a unique opportunity to study the effect of a sudden change in food

availability on reproductive success and population dynamics.

Data collection

We recorded reproduction data for active nests every other week for 8 years (November 2005–October 2013). We restricted data recording to a biweekly interval to reduce potential disturbances to a minimum. We collected the following data: number and condition of the eggs (incubated, abandoned or damaged), number, weight and estimated age of the nestlings, based on weight as well as plumage development after Johnston and Janiga (1995), and ring number on ringing. Laying date was defined according to the hatching date of the nestlings, which was calculated through their age based on weight. For abandoned or damaged clutches we assumed the laying date to be 9 days before egg recording, which represents half the incubation time of a successful incubation period of 18 days.

We made eight additional nocturnal breeding colony censuses during the study period. These censuses were necessary to identify all birds in the breeding colony and to determine the size of the colony. During the censuses, we closed the exit holes from the outside to hinder the pigeons from fleeing the loft. We caught all pigeons with hand nets, placed them in boxes and examined them individually. Numbered leg rings identified each bird and unmarked immigrant birds were ringed.

As nesting attempts are difficult to trace back to a definite date, we chose to rely on clutch interval length to draw reliable conclusions on the breeding history of a pair. The interval between successive clutches was defined as the number of days from the date the first egg was laid to the date the first egg of the next clutch was laid (Burley 1980). Clutch intervals were also calculated in cases when the second clutch ended in the loss of eggs or nestlings.

The number of breeding pairs nesting in the colony in a particular year was calculated as the number of nesting sites occupied at least once in that year, following Murton and Clarke (1968). The number of active nests per day was estimated using the data on recorded eggs, nestlings and fledged young. The number of pairs estimated to be breeding at the time of the nocturnal censuses was defined as the number of active nests with

eggs or nestlings from 1 week before until 1 week after each nocturnal census.

Hatching success was defined as the ratio of the number of nestlings hatched to the number of eggs laid, and fledging success as the ratio of fledged young to nestlings hatched (Dabert 1987). We had complete nest histories for all nests and thus expected no relevant nest exposure effect in our data because nests that were started and already failed between two of our biweekly intervals would still be recognized because the eggs were still present and were recorded. The mean annual reproductive success (ARS) was therefore calculated by dividing the number of fledged young by the number of breeding pairs in a particular year.

Data analysis

For clutch interval, a normal linear mixed model was fitted with log-transformed clutch interval as the response variable and port closure (before vs. after) as a binary categorical fixed effect. Nest and year were entered as random factors to account for non-independence within the data.

To model the change of the annual total number of nesting attempts, eggs, hatched and fledged young and the number of breeding pairs, we used Poisson models with a log-link to compare the means before and after port closure (one model for each of these five response variables and $n = 8$ for each of these models, as the response variables have one value per year). Models included an observation-level random effect to account for overdispersion. The number of nesting attempts per pair and per year, clutch size, and ARS were also estimated using Poisson models. For these three models, the measurement levels were the breeding pair ($n = 338$, for nesting attempts per pair and ARS) and the individual nesting attempt ($n = 831$, for clutch size), respectively. The factor 'before vs. after port closure' was fitted as a fixed categorical effect. Breeding pair (estimated as birds breeding in the same nest) and year (years from November to October the following year, as this study started in November) were fitted as random effects. There was no indication of overdispersion in these models.

To estimate the probability of hatching per egg (hatching success) and of fledging per hatchling (fledging success), two binomial models with the number of successes and failures were used (with a logit-link function). For both models, the

measurement level was the individual brood ($n = 831$ for hatching success and $n = 328$ for fledging success), and the number of successes and failures per brood were the response variable. Again, breeding pair and year were random effects and no strong overdispersion was observed.

Inference from these models used Bayesian 95% credible intervals based on 10 000 samples from the joint posterior distribution (the 95% Bayesian credible interval contains the true parameter value with a probability of 95%, given that the model assumptions are met; flat priors were used with the function `sim` of the R package `arm`).

RESULTS

Data from 831 broods collected in 8 years were included in the analysis. From 1429 eggs, 535 nestlings hatched (37.4%) of which 353 successfully fledged (66.0%). Overall ARS was 1.03 fledged young per breeding pair per year. The average number of Pigeons present during the nocturnal censuses was 111 in the years before and 82 in the years after port closure, representing a colony decrease of 26% after port closure.

Clutch interval

After port closure, average clutch interval was 37% longer than before, changing from 67 days (95% credible interval: 59–77) to 92 days (95% credible interval: 79–108).

Effect of food loss by port closure on breeding activity

Figure 1 shows reproduction parameters for the 4 years before and the 4 years after port closure. Table 1 shows the mean values for the years before and after port closure.

The number of breeding pairs per year fell significantly by 24.8% (95% Bayesian credible interval (BCI): –42.1%; –2.4%), and the number of nesting attempts per pair fell significantly by 28.5% (BCI: –38.0%; –17.4%) after port closure. However, mean clutch size did not change significantly. Overall, the number of eggs laid per year fell significantly after port closure by 45.6% (BCI: –63.1%; –20.0%).

There was no significant change in hatching success after port closure (4.5%; BCI: –22.3%;

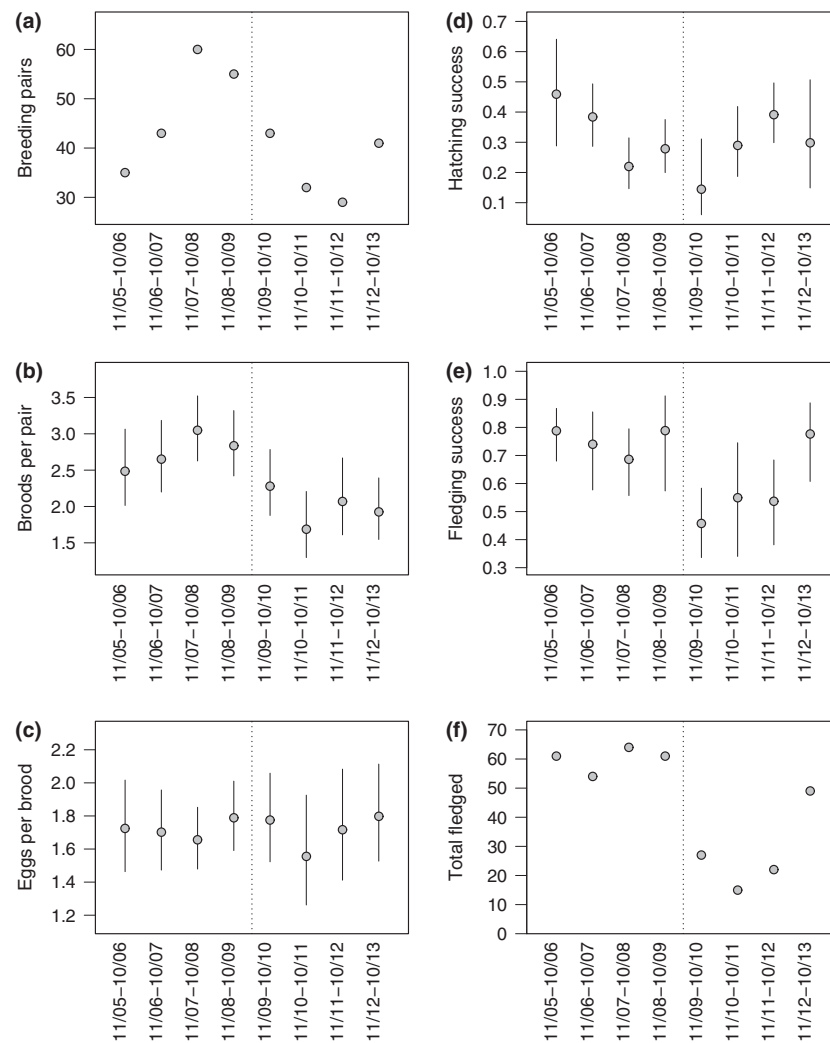


Figure 1. Subdivided annual reproduction parameters, indicated as November to October of the following year, of the Feral Pigeon population studied before and after a reduction in food loss. The vertical dashed line represents the closing of the Rhine Port St Johann, which is associated with significant food loss. Uncertainty is indicated by the vertical bar (95% Bayesian credible interval; exact values and their calculations are given in Table S1). (a) Breeding pairs, (b) broods per pair, (c) eggs per brood, (d) hatching success, (e) fledging success and (f) total fledged.

+40.8%), but fledging success decreased by 19.2% (BCI: -34.3%; -3.4%). The number of fledged young per pair per year (ARS), representing the accumulated impact of the port closure on the reproduction cycle up to the point of fledging, showed a marked reduction of 43.9% (BCI: -59.0%; -22.7%). Reduced fledging success of hatchlings combined with a reduced number of broods per pair after port closure are the main drivers of reduced overall reproductive success. In total, all of these effects combined to a significant reduction in the number of fledged young at the colony level of 54.6% (BCI: -68.9%; -33.2%) after

port closure. The summarized numbers of breeding pairs, eggs laid, nestlings and fledglings of 4 years before food shortage and 4 years under food loss are shown in Figure 2.

DISCUSSION

In our long-term study of 831 Feral Pigeon broods, we were able to show that food shortage following port closure affected reproductive success at several stages of the reproductive cycle, notably the number of breeding attempts made by pairs within a season and the survival rate of hatchlings to fledging.

Table 1. Mean values of breeding parameters before ($n = 4$ years) and after ($n = 4$ years) port closure, and the corresponding change (percentage change relative to the value before port closure; with 95% Bayesian credible intervals given in parentheses). Same model types as for Table S1 (see Supporting Information), but now including before vs. after port closure as a fixed factor: year and breeding pair were random factors where values are not sums per year.

	Measurement level	Before port closure	After port closure	% difference
Breeding pairs	Year ($n = 8$)	47.9 (40.2; 57.4)	36.1 (29.7; 43.8)	-24.8% (-42.1%; -2.4%)
Nesting attempts	Year ($n = 8$)	130 (101; 167)	71 (54; 93)	-45.5% (-62.2%; -21.5%)
Eggs	Year ($n = 8$)	223 (171; 293)	121 (92; 160)	-45.6% (-63.1%; -20.0%)
Nestlings	Year ($n = 8$)	84 (68; 105)	47 (38; 60)	-43.9% (-59.0%; -22.7%)
Fledglings	Year ($n = 8$)	60 (46; 77)	27 (20; 36)	-54.6% (-68.9%; -33.2%)
Nesting attempts per breeding pair	Breeding pair ($n = 338$)	2.74 (2.50; 3.00)	1.96 (1.73; 2.21)	-28.5% (-38.0%; -17.4%)
Clutch size	Nesting attempt ($n = 831$)	1.71 (1.61; 1.83)	1.73 (1.58; 1.89)	+0.8% (-9.6%; +12.4%)
Hatching success	Nesting attempt ($n = 831$)	31.8% (24.7%; 39.7%)	33.2% (25.3%; 42.2%)	+4.5% (-22.3%; +40.8%)
Fledging success	Hatched brood ($n = 328$)	72.4% (63.8%; 79.7%)	58.5% (47.4%; 68.9%)	-19.2% (-34.3%; -3.4%)
ARS (fledglings per breeding pair per year)	Breeding pair ($n = 338$)	0.83 (0.62; 1.13)	0.49 (0.35; 0.69)	-41.7% (-60.1%; -15.2%)

As the energetic costs for reproduction are higher for female than for male Feral Pigeons (Walsberg 1983), extra pressure is placed on females to find food. Pigeons in general have a long, sometimes even year-round, breeding season during which they lay multiple small clutches of almost exclusively two small eggs (Goodwin 1983, Robertson 1988). Once the eggs are laid, the energy expenditure of the incubation effort is minor. For both male and female pigeons, body weight increases during incubation accompanied by reduced food consumption, as the birds become relatively inactive while breeding (Riddle & Braucher 1934, Brisbin 1969). Clutches with only one egg, however, have a lower fledging rate than two-egg clutches (Johnston & Janiga 1995). This is because a singleton nestling may profit from more crop milk and food from both parents, but it suffers from the lack of reciprocal warming of another nestling. In Basel, the fledging success of one-egg clutches was about 10 times lower than that of two-egg clutches (Haag 1984).

The costs of reproduction are therefore relatively low up to the moment when the nestlings hatch. The number of young is thus limited by the energy needed to feed the nestlings. Energy requirements of two freshly hatched squabs with extremely high growth rates are difficult to meet, which consequently leads to a high mortality rate of the young, especially under food shortage conditions. In this study, the reduction in food caused by port closure resulted in a significantly larger number of the breeding pairs ceasing breeding. Pairs that continued to lay eggs produced significantly fewer clutches per year, with a significantly longer

average clutch interval. Clutch size remained largely stable, possibly due to the low energy input required from both parents, and hatching success was unaffected. Fledging success was reduced after port closure, possibly because the feeding of the young is the most energy-demanding part of the reproductive cycle for the food-stressed adults.

Overall, food shortage reduced the total number of fledged young by more than half, which consequently led to a smaller colony size. However, except for the number of broods per pair, all results showed an apparent increase of the studied reproduction parameters in the 4th year after port closure (Fig. 1). It is conceivable that the Pigeons initially suffered from the sudden food loss, but eventually managed to compensate over time by finding food elsewhere. Additionally, the declining colony size might also have positively influenced the studied reproductive parameters through density-dependent processes.

Our findings show that when adults invest in a brood after port closure, the more limited food resource conditions mainly affected the nestlings. While the investments in egg synthesis and incubation are minor, the energy demands of raising a brood with two nestlings are high (Riddle & Braucher 1934, Brisbin 1969, Walsberg 1983). Food scarcity seems to be surmountable during the first part of the reproductive cycle, but it intervenes strongly at hatching. A reduction in clutch size from two eggs to one egg, which would increase the amount of food available per squab, is not effective because a single hatchling is not able alone to meet its thermoregulation needs.



Figure 2. Illustration of the summarized number of breeding pairs, eggs laid, nestlings and fledglings in 4 years under normal food conditions (left, green columns) and 4 years under food loss (right, red columns). The ratio of the number of nestlings hatched to eggs laid is given as hatching success; the fledging success depicts the ratio of fledglings to nestlings (average values from a binomial mixed model with breeding pair and year as random effects and before vs. after port closure as fixed effects).

Our prediction that Feral Pigeon pairs would lay fewer clutches in a season under food shortage was confirmed. Our prediction that hatching success would decrease because pigeons have to spend more time on foraging was not confirmed. This observation can possibly be explained because the needs of the newly hatched young can be met easily by small amounts of crop milk, even when food availability is limited. However, the significantly reduced fledging success is the crucial factor in the reproductive cycle, strongly limiting the reproduction of Feral Pigeons under food limitation.

In conclusion, the energy demands of older nestlings may cause mortality during the rearing phase when food scarcity limits the energy intake of the foraging parental birds. Thus, the ability of a nestling to fledge depends on the amount of food available to the parent birds.

These results are relevant in terms of the management of Feral Pigeon populations in urban areas. Applying the results in practice suggests that food limitations mostly affect the young birds, as

more nestlings died after port closure. Feral Pigeon populations depend on abundant food for their maintenance and growth (Haag 1984). In urban areas, the available food base should not be supplemented by humans either intentionally or as human food waste. A more natural and ecological foraging behaviour would permit a healthier existence of Feral Pigeon populations in our cities, both for Pigeons and for people.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Table S1. Reproduction data and respective food situation per year. The first five data rows contain simple yearly sums. The other values are mostly estimates from Poisson models, for NI/Eg and FI/NI we used binomial models (number of successes vs. number of failures); in all cases, only an intercept was estimated. An observation-level random factor was included when overdispersion was indicated (residual deviance much larger than residual degrees of freedom, or using the function `dispersion_glm` from the R-package `blmeco` for mixed models). Where the value is per brood (Eg/BR, NI/BR, FI/BR), the breeding pair was included as a random term. Bayesian 95% credible intervals are given in parentheses. Abbreviations: normal = normal food situation, food loss = loss of main food source due to closing of the Rhine Port, BP = estimated number of breeding pairs, BR = broods, Eg = eggs, NI = nestlings, FI = fledglings, ARS (annual reproductive success) = fledglings per breeding pair. Standard deviations are given in parentheses.